

Kubernetes ReplicaSets:

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ minikube start
* minikube v1.38.1 on Microsoft Windows 11 Pro 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: default-storageclass, storage-provisioner
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ vim replicaset.yaml
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ ls
replicaset.yaml
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ cat replicaset.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: app1-rs
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.25
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl apply -f replicaset.yaml
replicaset.apps/app1-rs created
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
app1          1/1     Running   1 (7m58s ago)  21h
app1-rs-5zppm 1/1     Running   0           29s
app1-rs-9b7hw 1/1     Running   0           29s
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl delete pod app1-rs-5zppm
pod "app1-rs-5zppm" deleted from default namespace
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
app1          1/1     Running   1 (9m1s ago)  21h
app1-rs-9b7hw 1/1     Running   0           92s
app1-rs-fttk9 1/1     Running   0           9s
```

```

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assign
$ kubectl describe pod app1-rs-9b7hw
Name:          app1-rs-9b7hw
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Tue, 04 Aug 2026 11:18:06 +0530
Labels:        app=web
Annotations:    <none>
Status:        Running
IP:            10.244.0.12
IPs:
  IP:          10.244.0.12
Controlled By: ReplicaSet/app1-rs
Containers:
  nginx:
    Container ID:  docker://6a0a71dfb3f61bd68a52643bac627b098aef314d67eaa1e688c544190ef63c
    Image:         nginx:1.25
    Image ID:      docker-pullable://nginx@sha256:a484819eb60211f5299034ac80f6a681b06f89e6
    Port:          <none>
    Host Port:     <none>
    State:         Running
      Started:     Tue, 04 Aug 2026 11:18:21 +0530
    Ready:         True
    Restart Count: 0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-hgv68 (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers         True
  Initialized                       True
  Ready                             True
  ContainersReady                   True
  PodScheduled                      True
Volumes:

```

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$ vim replicaset.yaml

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$ ls
replicaset.yaml

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (ma
$ cat replicaset.yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: app1-rs
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.26

```

```

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods

```

NAME	READY	STATUS	RESTARTS	AGE
app1	1/1	Running	1 (13m ago)	21h
app1-rs-9b7hw	1/1	Running	0	5m59s
app1-rs-fttk9	1/1	Running	0	4m36s

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra
$ kubectl describe pod app1-rs-9b7hw
Name: app1-rs-9b7hw
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Tue, 04 Aug 2026 11:18:06 +0530
Labels: app=web
Annotations: <none>
Status: Running
IP: 10.244.0.12
IPs:
  IP: 10.244.0.12
Controlled By: ReplicaSet/app1-rs
Containers:
  nginx:
    Container ID: docker://6a0a71dfb3f61bd68a52643bac627b098aef314d67eaa1e688
    Image: nginx:1.25
    Image ID: docker-pullable://nginx@sha256:a484819eb60211f5299034ac80f6
    Port: <none>
    Host Port: <none>
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl delete pod app1-rs-fttk9
pod "app1-rs-fttk9" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
app1          1/1     Running            1 (14m ago) 21h
app1-rs-9b7hw 1/1     Running            0           7m30s
app1-rs-cz12z 0/1     ContainerCreating  0           6s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
app1          1/1     Running            1 (15m ago) 21h
app1-rs-9b7hw 1/1     Running            0           7m43s
app1-rs-cz12z 1/1     Running            0           19s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods app1-rs-cz12z
NAME          READY   STATUS             RESTARTS   AGE
app1-rs-cz12z 1/1     Running            0           70s
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl describe pod app1-rs-cz12z
Name: app1-rs-cz12z
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Tue, 04 Aug 2026 11:25:30 +0530
Labels: app=web
Annotations: <none>
Status: Running
IP: 10.244.0.15
IPs:
  IP: 10.244.0.15
Controlled By: ReplicaSet/app1-rs
Containers:
  nginx:
    Container ID: docker://8118e609e25ab29deb7f83365ee862b2d9f6a8246463a6b6bbbf0693df4f714
    Image: nginx:1.26
    Image ID: docker-pullable://nginx@sha256:41b194461e4bae16f9b25d68b0976ed4735b89ca625c89aad88e1c1c3
    Port: <none>
    Host Port: <none>
```

```
hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
app1-rs       2         2         2       10m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get rs app1-rs
NAME          DESIRED   CURRENT   READY   AGE
app1-rs       2         2         2       10m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl scale rs app1-rs --replicas=5
replicaset.apps/app1-rs scaled

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
app1          1/1     Running            1 (21m ago) 21h
app1-rs-9b7hw 1/1     Running            0           13m
app1-rs-cqfxx 1/1     Running            0           10s
app1-rs-cz12z 1/1     Running            0           6m16s
app1-rs-d4bbh 1/1     Running            0           10s
app1-rs-hzhnj 1/1     Running            0           10s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl scale rs app1-rs --replicas=2
replicaset.apps/app1-rs scaled

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
app1          1/1     Running            1 (21m ago) 21h
app1-rs-9b7hw 1/1     Running            0           14m
app1-rs-cz12z 1/1     Running            0           7m4s

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
app1-rs       2         2         2       14m
```

```

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl describe rs app1-rs
Name:          app1-rs
Namespace:     default
Selector:      app=web
Labels:        <none>
Annotations:   <none>
Replicas:      2 current / 2 desired
Pods Status:   2 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=web
  Containers:
    nginx:
      Image:          nginx:1.26
      Port:           <none>
      Host Port:      <none>
      Environment:    <none>
      Mounts:          <none>
  Volumes:            <none>
  Node-Selectors:     <none>
  Tolerations:        <none>
Events:
Type      Reason              Age   From                      Message
----      -
Normal    SuccessfulCreate     16m   replicaset-controller     Created pod: app1-rs-9b7hw
Normal    SuccessfulCreate     16m   replicaset-controller     Created pod: app1-rs-5zppm
Normal    SuccessfulCreate     15m   replicaset-controller     Created pod: app1-rs-fttk9
Normal    SuccessfulCreate     9m18s replicaset-controller     Created pod: app1-rs-cz12z
Normal    SuccessfulCreate     3m12s replicaset-controller     Created pod: app1-rs-cqfxx
Normal    SuccessfulCreate     3m12s replicaset-controller     Created pod: app1-rs-d4bbh
Normal    SuccessfulCreate     3m12s replicaset-controller     Created pod: app1-rs-hzhnj
Normal    SuccessfulDelete     2m21s replicaset-controller     Deleted pod: app1-rs-d4bbh
Normal    SuccessfulDelete     2m21s replicaset-controller     Deleted pod: app1-rs-hzhnj
Normal    SuccessfulDelete     2m21s replicaset-controller     Deleted pod: app1-rs-cqfxx

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
app1-rs-9b7hw 1/1     Running   0           18m   app=web
app1-rs-cz12z 1/1     Running   0           10m   app=web

```

```

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl delete rs app1-rs --cascade=orphan
replicaset.apps "app1-rs" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
app1-rs-9b7hw 1/1     Running   0           18m
app1-rs-cz12z 1/1     Running   0           11m

```

```

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get rs
No resources found in default namespace.

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kube ctl get pods
bash: kube: command not found

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
app1-rs-9b7hw 1/1     Running   0           22m
app1-rs-cz12z 1/1     Running   0           14m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl delete pod app1-rs-9b7hw
pod "app1-rs-9b7hw" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
app1-rs-cz12z 1/1     Running   0           15m

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl delete pod app1-rs-cz12z
pod "app1-rs-cz12z" deleted from default namespace

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get pods
No resources found in default namespace.

hp@Dharanidhar MINGW64 ~/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S_Assignment-2 (main)
$ kubectl get rs
No resources found in default namespace.

```